

WTW158 2022 Exam Multiple choice questions:**Each question is worth 1.5 marks.****Question 1:**The largest interval on which the function $f(x) = \sqrt{x-2} - \sqrt{3-x}$ is continuous is:

(a) $(-\infty, 2)$	(b) $(-\infty, 2]$	(c) $(-\infty, 3)$	(d) $(-\infty, 3]$	(e) $(2, 3)$	(f) $[2, 3]$	(g) None of (a)-(f)
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Question 2:

$$\lim_{x \rightarrow -\infty} \tan^{-1}(4x) =$$

(a) $-\infty$	(b) $-\frac{\pi}{2}$	(c) $-\pi$	(d) -2π	(e) 0	(f) None of (a)-(e)
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Question 3:The critical number(s) of $f(x) = \ln(x) - x$, is(are):

(a) $c = 1$	(b) $c = -1$	(c) $c = 1$ and $c = -1$	(d) $c = 0$	(e) f has no critical number	(f) None of (a)-(e)
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Question 4:The equation of the tangent line to $y = x^3 + 4$, in the point where $x = -1$, is:

(a) $y = 3x$	(b) $y = 3x + 6$	(c) $y = 3x - 6$	(d) $y = 3x^3 + 3x^2 + 1$	(e) $y = 3x^3 + 3x^2 - 1$	(f) None of (a)-(e)
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Question 5:

Consider the following statements:

1. $\ln(ab)^2 = 2 \ln a \ln b$
2. $\ln 3 = \ln 6 - \ln 2$
3. $\ln e^x = 1$

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

Question 6:

Consider the following statements:

1. If $f(x) = 3 \ln(x)$, then $f'(x) = \frac{3}{x}$
2. If $f(x) = \sinh(3x) + \cosh(3x)$, then $f'(x) = 3f(x)$
3. If $f(x) = e^{3x}$, then $f'(-x) = 3f(-x)$

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

Question 7:

$$\lim_{x \rightarrow \infty} x^2 e^{-x} =$$

(a) 0	(b) 1	(c) $-\infty$	(d) ∞	(e) None of (a)-(d)
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Question 8:

Consider the following statements:

1. The domain of $f(x) = \left(\frac{1}{2}\right)^x$ is $(0, \infty)$.
2. The domain of $f(x) = \ln(1 - x)$ is $(1, \infty)$.
3. The range of $g(x) = \ln(1 - x)$ is $(-\infty, \infty)$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

Question 9:

$$\int \frac{x^2}{x^2 + 1} dx =$$

- (a) $x^2 \tan^{-1} x + c$.
- (b) $x - \tan^{-1} x + c$.
- (c) $x + \tan^{-1} x + c$.
- (d) $\frac{x}{2} \ln(1 + x^2) + c$.
- (e) $2x \ln(1 + x^2) + c$.
- (f) None of these.

Question 10:

Suppose $\vec{v}$ is the vector with starting point $A(1,2,3)$ and endpoint $B(3,6,7)$, with length $|\vec{v}|$. Choose the correct option:

- (a) $\vec{v} = \langle 2, 4, 4 \rangle$ and $|\vec{v}| = 6$
- (b) $\vec{v} = \langle -2, -4, -4 \rangle$ and $|\vec{v}| = 6$
- (c) $\vec{v} = \langle 2, 4, 4 \rangle$ and $|\vec{v}| = \pm 6$
- (d) $\vec{v} = \langle -2, -4, -4 \rangle$ and $|\vec{v}| = \pm 6$
- (e) $\vec{v} = \langle 2, 4, 4 \rangle$ and $|\vec{v}| = \sqrt{10}$
- (f) $\vec{v} = \langle -2, -4, -4 \rangle$ and $|\vec{v}| = \pm \sqrt{10}$
- (g) None of the above is true.

Question 11:

Given: $h(x) \leq f(x) \leq \frac{\sin(x)}{2x}$ for all $x \in (\infty, 0) \cup (0, \infty)$. Consider the following statements:

1. If $\lim_{x \rightarrow 0} h(x) = 2$, then $\lim_{x \rightarrow 0} f(x) = 2$.
2. If $\lim_{x \rightarrow 0} h(x) = \frac{1}{2}$, then $\lim_{x \rightarrow 0} f(x) = \frac{1}{2}$.
3. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = 3$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

Question 12:

$$\lim_{x \rightarrow \infty} \frac{8x^3 + 5x^2 + 2x - 3}{1 - 2x^3} =$$

(a) -4	(b) 4	(c) $-\infty$	(d) ∞	(e) 0	(f) 8	(g) -8	(h) None of (a)-(f)
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SECTION B [Fill in questions] - 14 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.
Please do not scan your rough working or any empty pages.

SURNAME:

INITIALS:

STUDENT NUMBER:

SECTION B: Write only your final answer in the block provided. Only this will be marked.

QUESTION 13 [2 marks]

Evaluate the integral by interpreting it in terms of areas.

$$\int_0^3 (1 + \sqrt{9 - x^2}) dx =$$

QUESTION 14 [3 marks]

Find the derivative of the given function. Do not simplify your answer:

(a) $f(x) = e^x \ln x$

$f'(x) =$

(b) $f(x) = \cos(\cos x)$.

$f'(x) =$

(c) $f(x) = \frac{1}{\sqrt{x^2 + 1}}$

$f'(x) =$

QUESTION 15 [1 mark]

Find the value(s) of x such that the vectors $\langle 2x, 4, x \rangle$ and $\langle 3, 2, x \rangle$ are orthogonal.

QUESTION 16 [4 marks]

Determine the following integrals:

(a) $\int \frac{\cos(\ln t)}{t} dt =$

(b) $\int \frac{x}{x^2 + 1} dx =$

(c) $\int \frac{x}{\sqrt{x^2 + 1}} dx =$

(d) $\int 3^t \sec^2(3^t) dt =$

QUESTION 17 [2 marks]

The equation of a sphere is:

$$x^2 + y^2 + z^2 = 6x - 4y - 10z.$$

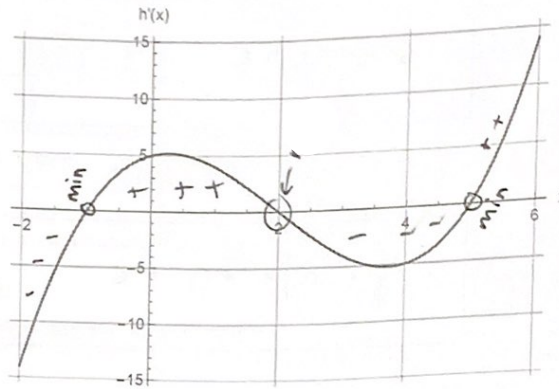
(a) The center of the sphere is:

(1)

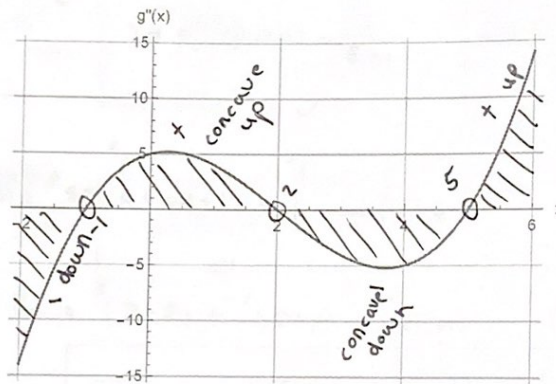
(b) The radius of the sphere is:

(1)

QUESTION 18 [2 marks]

(a) Consider the graph of h' :At which value(s) in its domain does the function h have (a) local maximum value(s)?

(1)

(b) Consider the graph of g'' :At which value(s) in its domain does the function g change concavity?

(1)

SECTION C [Full written answers] - 24 marks

QUESTION 19 [4 marks]

- (a) Complete the definition: The function f decreases on an interval I if: (1)

- (b) Given: $f(x) = x^3 \ln x$, $f'(x) = x^2(3 \ln x + 1)$ and $f''(x) = x(6 \ln x + 5)$.

- (1) Find the interval(s) on which the function f is decreasing. (2)

- (2) Find the interval(s) on which the function f is concave upward. (1)

QUESTION 20 [6 marks]

- (a) Use logarithmic differentiation to find $\frac{dy}{dx}$ if

(4)

$$y = \frac{3^x \tan 3x}{\sqrt{x^2 + x^3}}.$$

- (b) Find $\frac{dy}{dx}$ if $y = \tan^{-1}(x + y)$.

(2)

QUESTION 21 [4 marks]

Determine the following limits (if they exist):

(a) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$ (1)

(b) $\lim_{x \rightarrow 0} (1 + kx)^{\frac{1}{x}}, k > 0.$ (3)

QUESTION 22 [4 marks]

Determine $f(x)$ if f is differentiable, defined on the real line and:

(1) $f'(x) = ax^2 + bx$, with $a, b \in \mathbb{R}$.

(2) $f'(1) = 6$ and $f''(1) = 18$.

(3) $\int_1^2 f(x)dx = 18$

QUESTION 23 [3 marks]

Prove that the function f is continuous at a if it is differentiable at a . Explain all your steps.

QUESTION 24 [3 marks]

Assume that $\lim_{h \rightarrow 0} \frac{\cos(3h) - 1}{h} = 0$. Use the definition of a derivative to prove that

$$\frac{d}{dx} \sin(3x) = 3 \cos(3x).$$

Explain all your steps.